

Heterogeneous effects and causal forests

Status: draft for review

Intuition

An average effect can conceal differences across people, but identifying a conditional effect is harder than predicting an outcome. A causal forest learns local contrasts after nuisance adjustment under exchangeability and overlap. Honest splitting separates tree partition choices from local effect estimation. Independent evaluation reduces overfitting to apparent heterogeneity.

Model

$\tau(x) = E[Y(1) - Y(0) \mid X = x]$. An AIPW score is $g_1(X) - g_0(X) + D(Y - g_1(X))/m(X) - (1 - D)(Y - g_0(X))/(1 - m(X))$. Group averages of held-out scores estimate group ATEs under the maintained conditions. A best linear predictor projects the held-out score on an intercept and centered learned CATE; its slope describes calibration conditional on training fits.

Assumptions and interpretation

A pointwise CATE confidence interval describes uncertainty at a particular covariate vector. Averaging its endpoints does not yield a confidence interval for the average effect in a group, because covariance matters. A GATES analysis usually groups by out-of-sample estimated effect ranks; income quartiles define a different target. A BLP summarizes projection onto chosen effect predictors rather than proving a structural heterogeneity mechanism.

Application and limits

The forest is trained on one random half and evaluated on the held-out half. Income-quartile CATE means are descriptive. Held-out AIPW scores yield separate income-group ATE intervals with fixed clipping. These groups are not forest-ranked GATES, and their difference does not identify an effect of income. Heavy-tailed wealth, overlap and a single split remain fragile. No group is selected because it gives a desirable estimate.

Exam questions and answers

1. What makes forest splitting honest?

Answer. Partition selection and local effect estimation use separate information within the estimator.

2. Are averaged pointwise endpoints group confidence limits?

Answer. No; they omit the covariance needed for group-mean uncertainty.

3. What distinguishes income groups from GATES?

Answer. GATES groups are typically formed by held-out estimated-effect ranks rather than income alone.

4. Can a CATE pattern identify an income intervention?

Answer. No; it is treatment-effect heterogeneity conditional on income.

5. Why keep evaluation held out?

Answer. To avoid ranking and evaluating apparent heterogeneity on the same outcomes.

Reasoning exercise

A researcher averages the lower and upper pointwise forest interval endpoints within a high-income group and calls them a confidence interval for its mean effect. Why is that invalid? Answer: group-mean variance requires dependence across individual predictions; endpoint averaging does not supply it. Form income groups independently of outcome-based effect ranking and evaluate them on held-out observations. A held-out AIPW group score can provide a formal group estimand and influence-function uncertainty under the maintained nuisance conditions. This differs from GATES formed by estimated-effect ranks. Even valid treatment-effect heterogeneity by income does not identify the effect of changing income itself.

Verified reference

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